JOHN HOEKSEMA

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EXPERIENCE

Software Engineer | Fractal AI Accelerator 2026

- Created 20 repos with 400+ commits and 157 merged PRs over 650+ hours with Claude Code agents
- **Explore Mars**: 3D Mars globe (CesiumJS) built in 12 days, with 100+ monthly visitors; streams 21 GB of NASA terrain from S3 + CloudFront, loading sharper tiles on zoom; cut rover datapoints from 50,000 to 500
- **Sleeper Agents**: Replicated Anthropic's paper in 2 days; small models lost their backdoor in 1 epoch
- **Jar Jar bot**: Fine-tuned Qwen3-1.7B on a Mac to talk like Jar Jar Binks, then served it locally with Ollama

Software Engineering Intern | Stage Zero 2026

- Built MedScout (solo), a multi-agent pipeline that finds and ranks 50+ doctors per search for medical-device sales reps (web search, a research agent per doctor, rule-based scoring); a working rep tested it
- Built a market-sizing (TAM) agent that estimates a medical device's market bottom-up (procedures per year × cost per procedure), with tool use and a cited source for every number

Software Requirements Engineer | Gentex 2024 – 2026

- Wrote from scratch the requirements for configurable video software: one config object sets each automaker's hardware (7 categories) and error-state flowchart (3 states), with no software rewrite
- Software built from these requirements was delivered to BMW for testing
- Spearheaded versioned requirements so development ran in parallel, starting the move off waterfall

Software Analyst | Accenture Federal Services 2021 – 2022

- Built a Python decision-tree model on veteran data for the Department of Veterans Affairs; found missing phone and email best predicted low benefit use

PROJECTS

Biosignal Game Controller (EOG) 2026

- Built a real-time EOG pipeline (2 channels, 250 Hz) that reads eye movements from skin electrodes on a Cerelog board and moves a Pong paddle; modified its ESP32 firmware to reconfigure the ADS1299
- Organized and ran 3 public tournaments (8 players each, up to 30 spectators)
- Collected and published 237 recordings from 23 people, with a public web viewer

EDUCATION

B.S. Computer Science | University of Notre Dame 2021

SKILLS

- AI: Claude Code, context engineering, Anthropic API, agentic workflows, evals
- ML: LoRA, QLoRA, fine-tuning, MLX, llama.cpp, Ollama, Modal
- Software: Python, TypeScript, React, Next.js, Node.js, Express, WebSockets, CesiumJS, Tailwind CSS
- Data and cloud: SQL, Postgres, SQLite, Supabase, AWS S3, CloudFront, Vercel, Stripe, Git
- Hardware and signals: ESP32, ADS1299, BrainFlow, NumPy, SciPy, signal filtering